Physical Gibbs Certificates for Critical SK Quench Pairings

Odd-Sector Tails, Moving-Density Residuals, and Conditional Relaxation

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Abstract

We study the finite-observable interface in the October 5, 2026 preprint on universality of critical SK quench autocorrelations. The physical heat-bath generator, fair-spin initial law, Gibbs reference measure, critical edge coordinates, and order of limiting operations are retained. The main local result replaces a full two-copy density bound by its spin-odd component and gives an exact convolution-row certificate for the omitted critical modes. Semigroup factorization supplies a smaller warming prefactor. A moving-density residual estimate then controls finite-head propagation in the actual entrance law, with an explicit three-part error budget. Under the source's whole-space bounded-kernel and ergodicity premises, compactness gives separate even and odd relaxation rates for each fixed disorder realization; no uniform random gap is asserted. An exactly solvable two-spin Gibbs model checks the clock, density, and sector formulas. The Gaussian warming, sequential entrance selection, and Rademacher replacement inputs remain external to these local certificates. Contribution assessment: C3/5, a source-specific parity and moving-density certificate with conditional two-rate relaxation, not an independent proof of critical SK universality.

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1 Physical Model and Current Status

For fixed off-diagonal disorder $J = (J_{ij})_{i<j}$, put

$$\mu_n(\sigma) = Z_n^{-1} \exp\left(n^{-1/2} \sum_{i<j} J_{ij}\sigma_i\sigma_j \right), \qquad \sigma \in \{-1,1\}^n.$$

The inverse temperature is one and the external field is zero. Each site has a rate-one clock. On a ring at i, the new spin has conditional mean $\tanh h_i$, where $h_i = n^{-1/2}\sum_{j\neq i} J_{ij}\sigma_j$; retaining the old spin is an allowed mark. The positive generator and its form are

$$H_n f = \sum_i (\sigma_i - \tanh h_i) D_i f, \qquad D_i f = \tfrac{1}{2}(f(\sigma^{i,+}) - f(\sigma^{i,-})), \tag{1}$$
$$\langle f, H_n g\rangle_{\mu_n} = \mu_n \sum_i \operatorname{sech}^2(h_i) D_i f D_i g.$$

Thus $P_u = e^{-uH_n}$, with n expected rings per unit physical time. A one-ring-per-system convention changes the time normalization.

Let $T_n = n^{2/3}$, and let ν_n be independent fair spins, independent of J. The source observables are

$$A_n(t) = n^{-2/3}\sum_i \mu_n[\sigma_i P_{tT_n}\sigma_i], \tag{2}$$
$$B_n(s,t) = n^{-2/3}\sum_i \mathbb{E}_{\nu_n}[\sigma_i(sT_n)\sigma_i((s+t)T_n)].$$

These equal $n^{1/3}$ times the usual per-spin correlations. With an orthonormal eigenbasis u_a of the completed disorder matrix, set $x_{a,n} = n^{-1/3}\langle u_a, \sigma\rangle$. The independent GOE diagonal adds only a constant to the energy; it does not replace the physical spin generator. Orthogonality gives

$$\sum_{a=1}^{n} x_{a,n}(\sigma)x_{a,n}(\sigma') = n^{-2/3}\langle \sigma, \sigma'\rangle.$$

Writing $w_{n,s} = d(\nu_n P_{sT_n})/d\mu_n$, we obtain the exact representations

$$A_n(t) = \sum_a \mu_n[x_{a,n}P_{tT_n}x_{a,n}], \qquad B_n(s,t) = \sum_a \mu_n[w_{n,s}x_{a,n}P_{tT_n}x_{a,n}]. \tag{3}$$

The entrance density cannot be silently set equal to one.

Public status as of October 9, 2026. The assigned publisher preprint [1] claims Gaussian and Rademacher functional universality for these critical quench observables. This report neither endorses that global claim on the basis of finite tests nor identifies a source error. The earlier numerical equilibrium study of Billoire and Campbell [2] investigates critical $N^{-1/3}$ correlations on $N^{2/3}$ times; it is not a proof of quench universality. The current primary abstracts of Du and Huang [3, 4] state critical free-energy fluctuations and a random limiting overlap law, respectively. These are static results or preprint claims, not replacements for the dynamical entrance and history arguments here. Their full proofs were not read for this report.

2 Source Architecture and Retained Obligations

The source has ten scientific sections, of very unequal length. The following map distinguishes what the local results use from the much larger upstream argument.

Source module	Role and retained obligation
Introduction, background	Critical normalization, reflected GOE edge law, constrained Gibbs limit, and imported stationary form convergence.
Edge estimates	Spherical posterior caps, Haar comparison and class-uniform variance bounds; the long internal proof is only partly read.
Warming	Entropy and support profiles, actual physical kernels, and L^1-to-L^∞ bounds above a vanishing finite-n threshold.
Gaussian limits	Bounded resolvent optimizers, strong semigroup convergence, two-copy tails, and positive-time entrance densities.
Entrance selection	Sequential high-mode escape, causal Volterra inversion, localized stochastic coupling and uniqueness.
History and clock scores	Unnormalized survival histories, conditional interiors, predictable compensation and fourth-order gains.
Replacement, completion	Global cutoffs, Gaussian–sign interpolation, Taylor orders through 24, mesh estimates, and functional convergence.

For a fixed admissible edge sequence $\mathbf{g}$, the source uses $\ell_a = (g_a + b)^{-1} > 0$, with

$$\sum_a \ell_a = \infty, \qquad \sum_a \ell_a^2 < \infty.$$

The limit $\Pi_{\mathbf{g}}$ is a constrained product-Gaussian measure, obtained by disintegrating at a specified value of $\sum_a (Z_a^2 - \ell_a)$, not an unconditioned product law. Its cylinder densities contain the remaining centered-square density. The source imports or establishes, in its stated scope,

$$\Pi_{\mathbf{g}}|x_a|^j \le C_j \ell_a^{j/2}, \quad \mathcal{E}(F,F) = \Pi_{\mathbf{g}} \sum_a |\partial_a F|^2, \quad S_t = e^{-t\mathcal{A}}, \quad \mathcal{A} = c_* H_{\mathbf{g}}, \tag{4}$$

where the positive microscopic coefficient c_* must not be dropped. The form is closed, $\ker \mathcal{A}$ consists of constants, and the selected entrance densities satisfy

$$h_s \ge 0, \quad \Pi_{\mathbf{g}} h_s = 1, \quad h_{s+u} = S_u h_s, \quad \|h_s\|_\infty \le K(s), \quad s, u > 0.$$

The source warming bound has the form $K(s) = C \exp(C s^{-1.54})$, on its disorder-localized events. At finite n, its kernel bound is valid for $s \ge \varepsilon_n \downarrow 0$, not uniformly to time zero. For a fixed limiting environment the positive-time bounds give a nonnegative symmetric kernel $k_t \le K(t)$.

Two different tail quantities. At finite n, define

$$Q_{n,m} = \sum_{a>m} x_{a,n} \otimes x_{a,n}, \qquad V_{n,m} = \|Q_{n,m}\|^2_{L^2(\mu_n^{\otimes 2})} = \sum_{a,b>m} \mu_n [x_{a,n} x_{b,n}]^2. \tag{5}$$

The off-diagonal terms in the last sum cannot be discarded. In the limiting measure, independent coordinate sign symmetries give

$$V_m = \sum_{a>m} (\Pi_{\mathbf{g}} x_a^2)^2 \le C \sum_{a>m} \ell_a^2 \longrightarrow 0. \tag{6}$$

Consequently $Q = \sum_a x_a \otimes x_a$ exists in $L^2(\Pi_{\mathbf{g}}^{\otimes 2})$. This does *not* say that $x \in \ell^2$, or that $\sum_a \Pi_{\mathbf{g}} x_a^2$ is finite. The source's represented finite-n tail control is needed before passing to the infinite-mode limit.

Entrance and replacement are not automatic. Product-topology convergence to the zero configuration does not select the entrance law. The source proves a sequential escape condition involving $M(x) = \frac{1}{2}\sum_a x_a^2/(g_a + b)$ and $x_i^2/M(x)$, then uses a causal inversion and localized coupling. We retain that condition and that proof obligation. Likewise, a Gibbs scalar-product identity does not compare disorder laws. The replacement proof uses actual heat-bath likelihoods, normalized stationary endpoint denominators, independent conditional interiors, and compensation before restricting to future-dependent good events. It retains unnormalized survival and first-failure masses. We do not replace any of these mechanisms by a Gaussian surrogate.

3 An Odd-Sector Two-Copy Certificate

We state the local argument on a probability space (E, μ); both the finite spin model and the source limit fit its assumptions. Let $R : E \to E$ be a measure-preserving involution. On $L^2(\mu)$ write $Uf = f \circ R$ and $P_\pm = (I \pm U)/2$. Let $S_t = e^{-t\mathcal{A}}$ be a self-adjoint Markov semigroup commuting with U, with a symmetric nonnegative kernel k_t at the times used below. Let h_s be even probability densities with $h_{s+u} = S_u h_s$. Each real observable $x_a \in L^2(\mu)$ is odd. For a finite tail, or an $L^2(\mu^{\otimes 2})$ limit of such tails, write $Q_m = \sum_{a>m} x_a \otimes x_a$, $V_m = \|Q_m\|_2^2$.

Define convolution-row quantities, for almost every x, by

$$\begin{aligned} k_t^-(x,y) &= \tfrac{1}{2}\{k_t(x,y) - k_t(x,Ry)\}, \\ q_t(x) &= \int k_t(x,z)^2 \, d\mu(z), \qquad & c_t(x) &= \int k_t(x,z)k_t(Rx,z) \, d\mu(z), \\ \Gamma_-(s,t)^2 &= \frac{1}{2}\int h_s(x)^2\{q_t(x) - c_t(x)\} \, d\mu(x). \end{aligned} \tag{7}$$

These are definitions by integration of rows. A value such as $k_{2t}(x,x)$ of an arbitrary almost-everywhere kernel version is not defined on a null diagonal. Although q_t, c_t can be denoted by convolution versions of $k_{2t}(x,x)$ and $k_{2t}(x,Rx)$, no such pointwise choice is needed.

Theorem 3.1 (Sector tail and factorization)**.** *Under the preceding assumptions, suppose $h_s k_t \in L^2(\mu^{\otimes 2})$. Then*

$$\left|\langle Q_m, h_s k_t\rangle_{L^2(\mu^{\otimes 2})}\right| \le \Gamma_-(s,t)\sqrt{V_m}. \tag{8}$$

If $S_{2t} : L^1(\mu) \to L^\infty(\mu)$ has norm at most $K(2t)$, then

$$\Gamma_-(s,t) \le \sqrt{K(2t)}\,\|h_s\|_2 \le \sqrt{K(2t)K(s)} \tag{9}$$

whenever $h_s \le K(s)$. The same first bound holds for the full density norm $\|h_s k_t\|_{L^2(\mu^{\otimes 2})}$.

Proof. The kernel of $S_t P_-$ is k_t^-. Invariance, symmetry and commutation allow these identities after intersecting finitely many full-measure sets. Since Q_m is odd in each argument, projection in the second variable leaves it unchanged and replaces k_t by k_t^- in the pairing. The evenness of h_s also makes $h_s k_t^-$ odd in its first variable. Expansion of the square, followed by $z \mapsto Rz$, gives

$$\int |k_t^-(x,z)|^2 \, d\mu(z) = \tfrac{1}{2}(q_t(x) - c_t(x)).$$

Cauchy–Schwarz proves (8); it also passes to an L^2 tail limit.

For completeness, self-adjoint factorization gives

$$\|S_t\|_{2\to\infty}^2 = \|S_t S_t^*\|_{1\to\infty} = \|S_{2t}\|_{1\to\infty}.$$

Indeed the inequality from right to left follows from $\|S_t f\|_2^2 = \langle f, S_{2t} f\rangle \le \|S_{2t}\|_{1\to\infty}\|f\|_1^2$ for $f \in L^1 \cap L^2$, and duality; the other inequality follows by composition. For kernel operators on the present standard probability spaces the squared 2-to-∞ norm is the essential supremum of the squared row L^2 norm. This follows by taking a countable dense subset of the unit ball of L^2, so no exceptional set depends on a later choice of test function. Thus $q_t(x) \le K(2t)$ almost everywhere. Projection onto the odd sector contracts each row norm. Integration against h_s^2, and $\mu h_s^2 \le \|h_s\|_\infty \mu h_s$, prove the claim. □

The exact Γ_- removes the constant and all even modes of the transition kernel. Its improvement is not merely a change of coordinates: the target Q_m is orthogonal to those modes. The factorized warming cost is $\sqrt{K(2t)K(s)}$, rather than the crude $K(t)K(s)$. All of this still multiplies the *full equilibrium* V_m; there is no inference of that tail from a finite covariance block.

Corollary 3.2 (A narrower passage criterion)**.** *On a positive-time compact window, suppose all individual full densities $h_s k_t$, at finite n and in the limit, belong to their respective two-copy L^2 spaces. Suppose finite-head pairings converge, the exact sector norms $\Gamma_{-,n}$ are uniformly bounded, and $\lim_{m\to\infty} \limsup_{n\to\infty} V_{n,m} = 0$. Suppose also that the limiting $V_m \to 0$ and its sector norms are bounded there. Then the full pairings converge uniformly whenever the finite-head convergence is uniform on that window.*

Proof. Subtract the two finite-head pairings and bound the two remainders by (8). First fix m, pass $n \to \infty$, and then pass $m \to \infty$. The same three-term inequality holds after taking the supremum over the window. □

This criterion can use a direct two-copy norm certificate in place of a pointwise kernel bound for this particular passage. It does not dispense with warming elsewhere in the source. At finite n, (9) is used only once the fixed positive window lies above the source mesh threshold.

4 Residuals in the Evolving Physical Density

We next keep μ, S_t, h_s fixed as above. For real f, write $\|f\|_{h_s}^2 = \mu[h_s f^2]$. This is a seminorm if the density vanishes; all the following estimates remain valid in that case. The identity determining the correct weight is

$$\|S_r f\|_{h_s}^2 \le \mu[h_s S_r(f^2)] = \mu[(S_r h_s) f^2] = \|f\|_{h_{s+r}}^2. \tag{10}$$

The first step is Markov Jensen, not an assertion that S_r contracts every fixed weighted $L^2(h_s\mu)$ space.

Theorem 4.1 (Moving-density residual certificate)**.** *Fix $s, t > 0$, assume h_s is bounded, and fix a finite head $1 \le a \le m$. Let $v_a : [0,t] \to L^2(\mu)$ be strongly absolutely continuous, with $v_a(u) \in \mathrm{Dom}(\mathcal{A})$ almost everywhere and $\mathcal{A}v_a \in L^1([0,t]; L^2(\mu))$. Put*

$$e_a = x_a - v_a(0), \qquad r_a(u) = v_a'(u) + \mathcal{A}v_a(u), \qquad M_a(s) = \mu[h_s x_a^2].$$

Assume the weighted norms below are finite. Then

$$\left| \sum_{a\le m} \mu[h_s x_a \{S_t x_a - v_a(t)\}] \right| \le \mathcal{R}_m(s,t), \tag{11}$$

$$\mathcal{R}_m(s,t) := \sum_{a\le m} \sqrt{M_a(s)} \left(\|e_a\|_{h_{s+t}} + \int_0^t \|r_a(u)\|_{h_{s+t-u}}\, du \right).$$

All times in this statement are the semigroup times, including the physical factor T_n or c_* in $\mathcal{A}$.

Proof. The domain and integrability assumptions justify the variation of constants formula in $L^2(\mu)$:

$$S_t x_a - v_a(t) = S_t e_a - \int_0^t S_{t-u} r_a(u)\, du.$$

This identity also holds in the weighted seminorm by the continuous embedding from $L^2(\mu)$, since h_s is bounded. Minkowski and (10) bound its norm by the parenthesized expression. Weighted Cauchy–Schwarz against x_a and a finite sum prove (11). □

Proposition 4.2 (A complete three-part budget)**.** *Suppose $Q = \sum_a x_a \otimes x_a$ exists in $L^2(\mu^{\otimes 2})$, and the other assumptions of Theorems 3.1 and 4.1 hold. For any approximate density $\widehat{h}_s$ for which the expressions are defined, set*

$$\widehat{B}_m(s,t) = \sum_{a \le m} \mu[\widehat{h}_s x_a v_a(t)], \qquad F_m(x) = \sum_{a \le m} x_a(x) v_a(t,x).$$

If $F_m, h_s - \widehat{h}_s \in L^2(\mu)$, then

$$|B(s,t) - \widehat{B}_m(s,t)| \le \Gamma_-(s,t)\sqrt{V_m} + \mathcal{R}_m(s,t) + \|h_s - \widehat{h}_s\|_2 \|F_m\|_2. \tag{12}$$

Here $B = \langle Q, h_s k_t \rangle$; the same formula applies to the finite spin model with its finite tail.

Proof. Insert first the exact finite head, then its propagated approximations using the true density, and finally the approximate density. The three differences are respectively (8), (11), and $\mu[(h_s - \widehat{h}_s)F_m]$. Apply Cauchy–Schwarz to the last. □

There is no claim that a density estimate, residual estimate, or tail estimate is automatically available. Formula (12) states exactly what must be supplied. In particular it does not sum the factors $M_a(s)$ over all modes before taking a cutoff; that would reintroduce the divergent one-copy trace.

4.1 Physical frames and omitted residuals

The original Ram physical-metric and spectral-limit proofs [5, 6], freshly read for this report, provide the appropriate finite-space comparison. Let $J : \mathbb{R}^d \to L^2(\mu)$ be injective, with its range in $\mathrm{Dom}(\mathcal{A})$, and put

$$G = J^*J, \qquad P = JG^{-1}J^*, \qquad C = G^{-1}J^*\mathcal{A}J.$$

For $v(u) = Jc(u)$, its *full* residual is

$$\begin{aligned} r(u) &= J\{c'(u) + Cc(u)\} + (I - P)\mathcal{A}Jc(u), \\ \|r(u)\|^2_{L^2(\mu)} &= \|c'(u) + Cc(u)\|^2_G + \|(I - P)\mathcal{A}Jc(u)\|^2_2. \end{aligned} \tag{13}$$

The equality follows from orthogonality of P. Thus exact solution of the compressed ODE need not give zero physical residual. The second term is genuine omitted-space leakage, not a rounding error.

There is an additional caution in the present quench problem. Orthogonality for $L^2(\mu)$ is generally lost in $L^2(h_v\mu)$. For use in Theorem 4.1, either compute the full weighted residual, retain the cross term, or use

$$\begin{aligned} \|r\|_{h_v} &\le \|J(c' + Cc)\|_{h_v} + \|(I - P)\mathcal{A}Jc\|_{h_v}, \\ \|r\|_{h_v} &\le \sqrt{\|h_v\|_\infty}\, \|r\|_2. \end{aligned} \tag{14}$$

The first norm has physical Gram matrix $G_v = J^* M_{h_v} J$, not the Euclidean identity. The last inequality is an alternative bound on the full residual, not a bound on the preceding sum by the same right-hand side. Original finite-moment perturbation arguments [7] likewise require verified frame and quadrature margins. A well conditioned coefficient matrix alone certifies neither G_v nor the omitted term in (13).

Practical order of use. Fix a positive-time window and a finite head. Certify its actual Gibbs moment matrices and full residuals in the moving weights. Supply a density error against the same measure, and separately bound the infinite or finite-n two-copy tail. Only then use (12). This permits improved residuals to be useful without claiming that a finite approximation constructs the entrance law or establishes Gaussian–Rademacher universality.

5 Conditional Two-Rate Relaxation

The source proves relaxation by $h_s \to 1$ in L^2, then finite-head and tail arguments. The same source premises imply a sharper fixed-environment statement. This is a consequence of classical compact self-adjoint semigroup theory, used here with the two different parity sectors of the quench pairing.

Let $\mathscr{H}_+ = \{f : Uf = f,\ \mu f = 0\}$ and $\mathscr{H}_- = \{f : Uf = -f\}$. We state the result when both sectors are nonzero; a zero sector contributes zero and needs no rate.

Theorem 5.1 (Quenched parity gaps and relaxation)**.** *Assume S_τ has a bounded kernel for every $\tau > 0$, $\ker \mathcal{A} = \mathbb{R}\mathbf{1}$, and the involution assumptions above. There are numbers $\lambda_+, \lambda_- > 0$, depending on the entire operator, such that*

$$\|S_u|_{\mathscr{H}_\pm}\|_{2\to 2} \le e^{-\lambda_\pm u}.$$

Suppose also that $Q \in L^2(\mu^{\otimes 2})$ with $V_0 = \|Q\|_2^2$, and the even entrance densities obey the stated semigroup relation. For $s \ge s_0 > 0, t \ge t_0 > 0$,

$$|B(s,t) - A(t)| \le \sqrt{V_0 K(2t_0)}\, \|h_{s_0} - 1\|_2 e^{-\lambda_+(s-s_0)-\lambda_-(t-t_0)}. \tag{15}$$

One also has

$$\Gamma_-(s,t)\sqrt{V_m} \le \sqrt{V_m K(2t_0)}\, \|h_s\|_2 e^{-\lambda_-(t-t_0)}. \tag{16}$$

Proof. A bounded kernel on a probability space is square integrable. Hence S_τ is Hilbert–Schmidt and compact. Self-adjointness, positivity of $e^{-\tau\mathcal{A}}$, and $\ker \mathcal{A} = \mathbb{R}\mathbf{1}$ imply that the eigenvalue one is simple. If the norm on either closed invariant nonconstant sector were one, compactness would furnish an eigenvector with eigenvalue one in that sector, a contradiction. Spectral calculus then gives a strictly positive lower spectral edge $\lambda_\pm$ for $\mathcal{A}$ on that sector, and the displayed norm bound for all $u \ge 0$.

For almost every x, the row $k^-_{t_0}(x,\cdot)$ belongs to $\mathscr{H}_-$. Kernel composition in L^2 gives

$$k^-_t(x,\cdot) = S_{t-t_0} k^-_{t_0}(x,\cdot), \qquad \|k^-_t(x,\cdot)\|_2^2 \le e^{-2\lambda_-(t-t_0)} K(2t_0).$$

This is again an integrated-row identity; it uses no arbitrary diagonal version. Multiply by h_s^2 to obtain (16). For the difference $B - A$, multiply instead by $(h_s - 1)^2$, pair with Q, and apply Cauchy–Schwarz. Since $h_s - 1 = S_{s-s_0}(h_{s_0} - 1)$ belongs to $\mathscr{H}_+$, its norm is at most $e^{-\lambda_+(s-s_0)}\|h_{s_0} - 1\|_2$. This proves (15). □

The actual limiting SK measure has global sign symmetry and its gradient form commutes with that symmetry. Fair finite-spin entrance densities are even; this property passes through the source's bounded cylinder pairings. Thus the theorem applies *conditionally on* the source whole-space construction and kernel bounds, with $\mathcal{A} = c_* H_{\mathbf{g}}$. It does not produce a uniform gap over edge environments, a new critical exponent, or a finite-n uniform mixing theorem.

Proposition 5.2 (What a quantitative gap certificate needs)**.** *Let*

$$R_-(\tau) = \|S_\tau P_-\|_{\mathrm{HS}}^2, \qquad R_+(\tau) = \|S_\tau (P_+ - \mathbf{1}\otimes\mathbf{1})\|_{\mathrm{HS}}^2.$$

If a certified whole-space upper bound is $R_\pm(\tau) \le r < 1$, then

$$\lambda_\pm \ge -\frac{\log r}{2\tau}. \tag{17}$$

For either nonzero sector some sufficiently large τ makes $R_\pm(\tau) < 1$, but the assertion is not effective without further information.

Proof. The Hilbert–Schmidt sum is $\sum_j e^{-2\tau\lambda_j}$, which is at least its largest summand $e^{-2\tau\lambda_\pm}$. For eventual smallness, fix $\tau_0 > 0$, use summability at τ_0, and apply dominated convergence as $\tau \to \infty$. □

Finite eigenvalue lists give lower bounds on this trace, not the needed upper bound. A discarded-space certificate is essential. For example, the symmetric two-state chain with jump rate $\varepsilon > 0$ has kernel bounded by two at every positive time, yet its nonconstant gap is 2ε. Therefore a bound $K(t) \le 2$ alone gives no uniform positive gap. The sector traces above are traces of a smoothed operator, not the divergent coordinate quantity $\sum_a \mu x_a^2$.

6 An Exact Two-Spin Gibbs Example

Take $n = 2$, put $\kappa = J_{12}/\sqrt{2}$ and $a = \tanh\kappa \in (-1, 1)$. With $z = \sigma_1\sigma_2$,

$$\mu(\sigma) = \frac{1 + az}{4}, \quad H\sigma_1 = \sigma_1 - a\sigma_2, \quad H\sigma_2 = \sigma_2 - a\sigma_1, \quad H(z-a) = 2(z-a).$$

The constant, even mean-zero, and two odd eigenfunctions are $1, z-a, y_+ = (\sigma_1+\sigma_2)/2, y_- = (\sigma_1-\sigma_2)/2$, with eigenvalues $0, 2, 1-a, 1+a$. Their norms are

$$\mu(z-a)^2 = 1 - a^2, \qquad \mu y_\pm^2 = (1 \pm a)/2.$$

This is the actual heat-bath generator (1). No Gaussian variable has been substituted for a spin.

Let $S = s\,2^{2/3}$ and $L = t\,2^{2/3}$ be physical waiting and lag times. A fair start has $\mathbb{E}z(0) = 0$, hence

$$\mathbb{E}z(S) = a(1 - e^{-2S}), \qquad h_s = 1 - \frac{ae^{-2S}}{1-a^2}(z-a). \tag{18}$$

The second formula follows either by dividing the resulting four-state law by μ, or by propagating its even mean-zero density. In particular h_s is not the constant one. Choose the two normalized edge coordinates $x_\pm = 2^{-1/3}(\sigma_1 \pm \sigma_2)/\sqrt{2} = 2^{1/6} y_\pm$. The ordering of the two labels is immaterial to the full trace. Their exact quench moments are

$$\mu[h_s x_\pm^2] = 2^{-2/3}\{1 \pm a(1 - e^{-2S})\}.$$

Consequently

$$B(s,t) = 2^{-2/3}\big[(1+a(1-e^{-2S}))e^{-(1-a)L} + (1-a(1-e^{-2S}))e^{-(1+a)L}\big], \tag{19}$$

$$B(s,t) - A(t) = -2^{-2/3}ae^{-2S}\{e^{-(1-a)L} - e^{-(1+a)L}\}. \tag{20}$$

Thus the even waiting-time rate is two and the slow odd lag rate is $1-|a|$, in physical units. In critical rescaled units they are multiplied by $2^{2/3}$. The two rates can be very different.

The row certificate is explicit. Write $e = z - a$ and $v_\pm = (1 \pm a)/2$. The kernel relative to μ is

$$k_L(x,y) = 1 + e^{-2L}\frac{e(x)e(y)}{1-a^2} + e^{-(1-a)L}\frac{y_+(x)y_+(y)}{v_+} + e^{-(1+a)L}\frac{y_-(x)y_-(y)}{v_-}. \tag{21}$$

The odd projection deletes exactly the first two terms. For example, retaining only y_+ leaves $Q_{\text{tail}} = y_- \otimes y_-$, $V_{\text{tail}} = v_-^2$, and the exact omitted contribution is $e^{-(1+a)L}\mu[h_s y_-^2]$. Its absolute value is bounded by $\Gamma_-(s,L)v_-$, as predicted. This gives a nonzero-density, nonstationary check of the sector certificate rather than merely a check at equilibrium.

Sharpness and limitations of the example. It verifies the two distinct decay mechanisms and the physical time factor, not equality in every Cauchy–Schwarz bound. The approach $a \to 1$ exhibits a slow odd mode while the even gap remains two. It is a finite Gibbs example, not an approximation theorem for the critical infinite-n SK model. For exact rational tests we use $a = k/d$, $S = d\log b$, and $L = d\log c$, with integers $b, c > 1$. All eigenvalue exponentials are then rational: $e^{-2S} = b^{-2d}$ and $e^{-(1\mp a)L} = c^{-(d\mp k)}$. The irrational common critical-coordinate factor is kept symbolic; tests use the rational $y_\pm$ basis.

7 Advantages, Costs, and Further Work

Demonstrated increment. The source already has the essential two-copy tail and a correct weak-density/strong-semigroup passage. We do not claim those as new. The local increment is an exact odd-sector row certificate, its factorized warming cost, and a complete residual budget in the true evolving Gibbs density. Coupling these with compactness gives the two-rate estimate (15), beyond the source's qualitative relaxation formulation. The proofs use classical Markov Jensen, Hilbert-space duality, symmetry projection, Duhamel and compact spectral theory; these tools are not new Ram inventions.

Advantages. The sector projection removes irrelevant equilibrium and even kernel components without changing the observable. The moving weight can be much less costly than a supremum-density majorant when a residual is concentrated away from entrance mass. The finite-head/infinite-tail separation remains valid even though the one-copy coordinate trace diverges. The residual formula is invariant under a change of finite synthesis basis provided its physical Gram matrices are transformed as well. The separate waiting and lag rates explain which sector a numerical or analytic gap certificate must control.

Costs and nonclaims. Obtaining $V_{n,m}$, a full residual, or a whole-space trace upper bound may be harder than solving the compressed system. None is furnished by small coefficient residuals alone. The source's kernel constants and disorder events are inherited; no measured computational speedup is asserted. The sector gaps are environment-dependent, and an annealed exponential bound would require additional control of their distribution and of the prefactors. Uniform bounds do not extend automatically to $s = 0$ or $t = 0$. The two-copy method permits a positive-lag trace, not an unregularized infinite coordinate trace. A finite kernel calculation neither proves high-mode escape nor justifies replacing fair-spin quench histories by stationary ones. All conditional copy laws, predictable compensators, and fourth-order replacement estimates remain genuine source obligations.

Feasible extensions. A rigorously enclosed odd-row norm could replace the coarser $K(2t)$ in (9). If the finite head is chosen adaptively, its discarded covariance must still be certified uniformly under the same disorder law. For several odd observables, a positive matrix of their two-copy inner products gives a joint certificate by applying the same Hilbert-space argument to linear combinations. A broken spin-flip symmetry would require retaining the even component rather than applying the present cancellation. For approximate entrance densities not satisfying an exact semigroup relation, their density-equation residual must enter an additional Duhamel term; (10) applies to the true density, not automatically to its approximation. An effective bound for a parity trace with a certified discarded-space tail is a concrete route toward quantitative rates, but is not supplied here.

8 Evidence and Contribution Boundary

The source-read ledger records pre-read and post-read SHA-256 hashes, each bounded literal display and its unread complement. The following is an author scope, not the parent's independent review scope. All of the introduction, background, Gaussian limits, entrance selection, history cutoffs, clock scores, replacement, completion, main file, bibliography and README were read. The much longer edge and warming proofs were scoped as follows:

File	Literal lines read	Unread proof lines
03-edge-estimates.tex	1–270	271–1895
04-warming.tex	1–120, 1174–1623	121–1173
Original physical-metrics chapter	1–210	none in text
Original spectral-limits chapter	1–242	none in text
Original TR248	349–457	1–348, 458–1302
Later core16 locator	none	1–939

TR248 line 349 begins at the end of an earlier display; only the complete later statements and proofs within the recorded interval support the frame-perturbation comparison. Referenced external figures and the full confidential corpus were not visually or exhaustively audited. The companion stationary universality preprint was an imported input, not a fresh proof read. Public current literature reads and their abstract-only limits are recorded separately.

The owned exact-check script constructs rational four-state Gibbs measures and physical heat-bath generators, their spectral kernels, nonconstant entrance densities, parity projections and weighted Jensen fixtures. It checks exact equalities and squared inequalities, not floating-point approximations to limiting SK laws. Parameter-case and assertion counts appear in its receipt; the script and receipt are frozen with this report. The parent review and parent tests are not counted as author tests. No foreign checker is replayed.

The assessment is C3 for the combined source-specific certificate and conditional relaxation increment, not for elementary identities in isolation. Applicability remains L3. No new probabilistic input, independent global universality proof, verified random spectral-gap distribution, or improved critical scaling exponent is claimed. No source mathematical correction is asserted; its qualitative relaxation and conservative two-copy bound remain valid.

References

[1] OpenAI, *Universality of critical quench autocorrelations in the Sherrington–Kirkpatrick model*, October 5, 2026, captured publisher preprint. Public source directory. The local captured source, not live branch status, is the object of comparison.

[2] A. Billoire and I. A. Campbell, *Dynamics in the Sherrington–Kirkpatrick Ising spin glass at and above T_g*, Physical Review B **84** (2011), 054442. arXiv:1105.1902; publisher DOI. Public abstract/version metadata and the returned beginning through the critical scaling discussion of v1 were read; the full simulation study was not audited.

[3] H. Du and B. Huang, *Fluctuations of the Sherrington–Kirkpatrick free energy at critical temperature*, arXiv:2607.02172v2, July 8, 2026. Primary abstract and version metadata read.

[4] H. Du and B. Huang, *Overlap distribution of the critical Sherrington–Kirkpatrick model*, arXiv:2608.08752v1, August 9, 2026. Primary abstract and version metadata read.

[5] M. Subbarao, original confidential RamCalBooks, volume 21, `ch01d-physical-metrics.tex`, lines 1–210. Physical synthesis Gram matrices and full residual splitting; fresh local text read, no online disclosure.

[6] M. Subbarao, original confidential RamCalBooks, volume 21, `ch01f-spectral-limits.tex`, lines 1–242. Graph-domain, omitted-space and finite-section limitations; fresh local text read.

[7] M. Subbarao, original confidential RamFoundationTRs, `TR-rev-1-old-tr248.tex`, lines 349–457. Complete finite frame/perturbation statements within this interval were used; the rest of the manuscript was not read.